

PROGRAM

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#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h> // Required for sleep()

#define BUFFER_SIZE 5

sem_t empty;
sem_t full;
pthread_mutex_t mutex;
int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;

void *producer(void *arg) {
    int item;
    // Infinite loop for continuous production
    while(1) {
        item = rand() % 100; // Produce random item

        sem_wait(&empty); // Wait for empty slot
        pthread_mutex_lock(&mutex); // Enter Critical Section

        // Critical Section: Add item to buffer
        buffer[in] = item;
        printf("Producer produced: %d\n", item);
        in = (in + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex); // Exit Critical Section
        sem_post(&full); // Signal that a slot is full

        sleep(1); // Sleep for 1 second to make output readable
    }
}

void *consumer(void *arg) {
    int item;
    // Infinite loop for continuous consumption
    while(1) {
        sem_wait(&full); // Wait for filled slot
        pthread_mutex_lock(&mutex); // Enter Critical Section

        // Critical Section: Remove item from buffer
        item = buffer[out];
        printf("Consumer consumed: %d\n", item);
        out = (out + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex); // Exit Critical Section
        sem_post(&empty); // Signal that a slot is empty

        sleep(1); // Sleep for 1 second to make output readable
    }
}
```

```

int main() {
    pthread_t prod_tid, cons_tid;

    // Initialize semaphores
    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    pthread_mutex_init(&mutex, NULL);

    printf("Producer-Consumer Simulation Started (Press Ctrl+C to stop)...\n");

    // Create threads
    pthread_create(&prod_tid, NULL, producer, NULL);
    pthread_create(&cons_tid, NULL, consumer, NULL);

    // Wait for threads (Technically unreachable due to while(1))
    pthread_join(prod_tid, NULL);
    pthread_join(cons_tid, NULL);

    // Destroy semaphores
    sem_destroy(&empty);
    sem_destroy(&full);
    pthread_mutex_destroy(&mutex);

    return 0;
}

```

OUTPUT

```

blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_5$ gcc 5.c
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_5$ ./a.out
Producer-Consumer Simulation Started (Press Ctrl+C to stop)...
Producer produced: 83
Consumer consumed: 83
Producer produced: 86
Consumer consumed: 86
Producer produced: 77
Consumer consumed: 77
Producer produced: 15
Consumer consumed: 15
Producer produced: 93
Consumer consumed: 93
Producer produced: 35
Consumer consumed: 35
Producer produced: 86
Consumer consumed: 86
Producer produced: 92
Consumer consumed: 92
Producer produced: 49
Consumer consumed: 49
^C
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_5$

```